
MBA • DATA ANALYTICS CAPSTONE

Customer Segmentation & Shopping Trends Analysis

Powered by Power BI • SQL • Python • Machine Learning

POWER BI

SQL

PYTHON

MACHINE LEARNING

PREPARED BY

Data Analytics Team

June 2026

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Business Problem

Why segmentation matters

02

Dataset

3,900 customer records

03

Objectives

8 analytical goals

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Tools

Power BI, SQL, Python, ML

05

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Quality preparation

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KMeans clustering

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Conclusion

Outcomes & roadmap

Mass marketing is dead. Customers expect personalised experiences — and retailers face shrinking margins.

1

Generic Campaigns

One-size-fits-all marketing yields <1% conversion vs 4-8% for targeted campaigns.

2

Discount Erosion

Untargeted promotions destroy margin without lifting loyalty.

3

Inventory Mismatch

Without seasonal/segment forecasts, stock-outs and dead-stock co-exist.

4

Churn Blind Spot

No view of who is loyal, occasional, or at risk.

5

Decision Latency

Decisions made on intuition, not on behavioural evidence.

Why it matters

80%

of revenue from 20% of customers

5x

cheaper to retain than acquire

60%

lift from segmented campaigns

3x

ROI on personalisation

Project Objectives

Eight analytical goals that frame the deliverable

04 / 40

1

Discount Opportunities

Quantify revenue lift from targeted promotions

2

Customer Spending

Measure average ticket and lifetime value patterns

3

Season Analysis

Identify peak periods for inventory & marketing

4

Location Analysis

Rank states by revenue contribution

5

Age Analysis

Profile spending behaviour by age cohort

6

Revenue Analysis

Decompose total revenue by category & channel

7

Customer Segmentation

Apply ML to create actionable clusters

8

Marketing Recommendations

Translate insights into a campaign playbook

Dataset Overview

Customer Shopping Trends — 3,900 records, 19 attributes

RECORDS 3,900 customer transactions	ATTRIBUTES 19 behavioural & demographic	CATEGORIES 4 Clothing, Accessories, ...	REGIONS 50 US states covered
--	--	--	---

? Customer ID	? Age	♀ Gender	? Item Purchased	? Category
? Purchase Amount	? Location	? Size	? Color	? Season
? Review Rating	? Subscription	? Payment Method	? Shipping Type	? Discount
? Promo Code	? Previous Purchases	? Preferred Payment	? Frequency	

Dataset Schema

Column definitions and analytical role

Column	Type	Description	Role
Customer_ID	Integer	Unique customer identifier	Primary Key
Age	Integer	Customer age (18-70)	Demographics
Gender	Text	Male / Female	Demographics
Item_Purchased	Text	SKU-level product	Transaction
Category	Text	Clothing, Accessories, Footwear, Outerwear	Product
Purchase_Amount_USD	Decimal	Transaction value in USD	Revenue
Location	Text	US state	Geography
Size / Color	Text	Product variants	Product
Season	Text	Spring / Summer / Fall / Winter	Time
Review_Rating	Decimal	Customer rating 1-5	Satisfaction
Subscription_Status	Text	Yes / No	Loyalty
Payment_Method	Text	Card / Cash / Digital	Channel
Discount_Applied	Text	Yes / No	Promotion
Previous_Purchases	Integer	Count of past purchases	Loyalty
Frequency_of_Purchases	Text	Weekly / Monthly / etc.	Engagement

SQL

DATABASE QUERYING

- Joins, GROUP BY, CTEs
- Window functions
- Ranking & aggregation
- Source-of-truth queries

Python

ANALYSIS & ML

- Pandas — data wrangling
- NumPy — vector math
- Matplotlib — EDA charts
- Scikit-Learn — clustering

Power BI

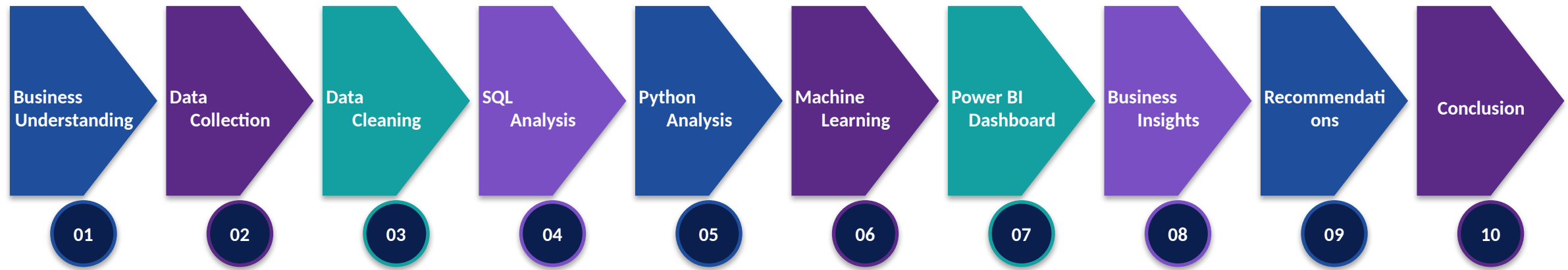
VISUALISATION

- Power Query — ETL
- DAX — measures
- Star-schema model
- Interactive dashboard

Machine Learning

SEGMENTATION

- KMeans clustering
- Elbow & silhouette
- Feature scaling
- 3-cluster solution



CRISP-DM inspired pipeline

Each phase produces a versioned artefact: cleaned dataset → SQL views → Python notebook → trained model → published Power BI report. The output of each stage is the input to the next, ensuring traceability.



Missing Values

Check NaNs across all 19 columns. None found in critical fields; defensive imputation prepared for production load.



Duplicates

Verify unique Customer_ID; 0 duplicate rows detected.



Data Types

Cast Age & Previous_Purchases → int, Purchase_Amount_USD → float, Review_Rating → float, dates as datetime.



Feature Engineering

Created Age_Group (5 buckets), Discount_Flag, High_Value flag, Revenue_Bucket.



Encoding

Label encoding for Gender; one-hot for Category, Season, Payment_Method to feed clustering.

-- Top categories by revenue

```
SELECT Category,
       COUNT(*)           AS txns,
       SUM(Purchase_Amount_USD) AS revenue,
       AVG(Purchase_Amount_USD) AS avg_ticket
FROM customer_transactions
GROUP BY Category
ORDER BY revenue DESC;
```

-- Cluster ranking with window function

```
WITH cluster_rev AS (
    SELECT cluster, SUM(Purchase_Amount_USD) rev
    FROM customer_transactions GROUP BY cluster )
SELECT cluster, rev,
       RANK() OVER (ORDER BY rev DESC) AS rk
FROM cluster_rev;
```

GROUP BY

Aggregate revenue per category, season, location

CASE WHEN

Bucket customers into Age Groups & High-Value flags

CTE (WITH)

Layer multi-step logic for clarity and re-use

WINDOW FN

Rank top customers, percentile, running totals

RANK()

Top-N customers / states by revenue

AGGREGATION

SUM, AVG, COUNT — KPIs feeding Power BI

```
import pandas as pd, numpy as np
import matplotlib.pyplot as plt, seaborn as sns
```

```
df = pd.read_csv('shopping_trends.csv')
df.info(); df.describe()
```

```
# revenue by category
rev = (df.groupby('Category')
       ['Purchase_Amount_USD']
       .agg(['sum', 'mean', 'count']))
```

```
# correlation matrix
corr = df[num_cols].corr()
sns.heatmap(corr, annot=True)
```

```
# segment-level visual
sns.boxplot(x='cluster',
            y='Purchase_Amount_USD', data=df)
```

Pandas

Wrangling, group-by, reshape & merge. Backbone of every transformation.

EDA

Distribution checks, outlier scan, correlation matrix, violin & box plots.

Correlation

Identified weak linear ties — a signal that clustering (non-linear) is the right tool.

Grouping

Revenue, ticket, frequency by Category, Age, Gender, Season, Location.

Visualisation

Matplotlib + Seaborn — every claim in this deck is backed by a chart.

Clustering is unsupervised — perfect for finding structure when no label exists.

Why Clustering



Customers are not pre-labelled. Clustering surfaces hidden behavioural cohorts.

KMeans



Partitions customers into k groups by minimising within-cluster variance on scaled features.

Elbow Method



Plot inertia vs k. The 'elbow' at k=3 indicates diminishing returns beyond 3 clusters.

Silhouette Score



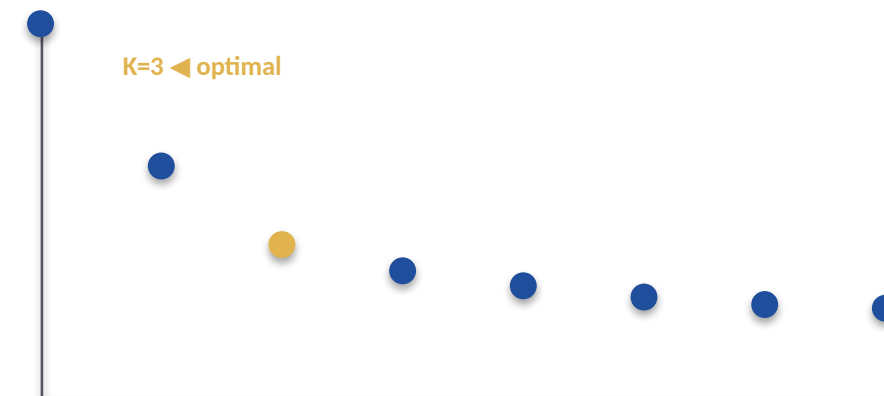
Validates separation — peak silhouette at k=3 (~0.42), confirming the elbow.

Why K=3

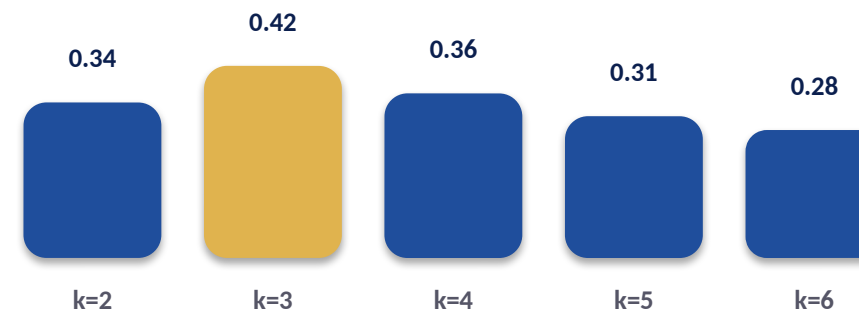


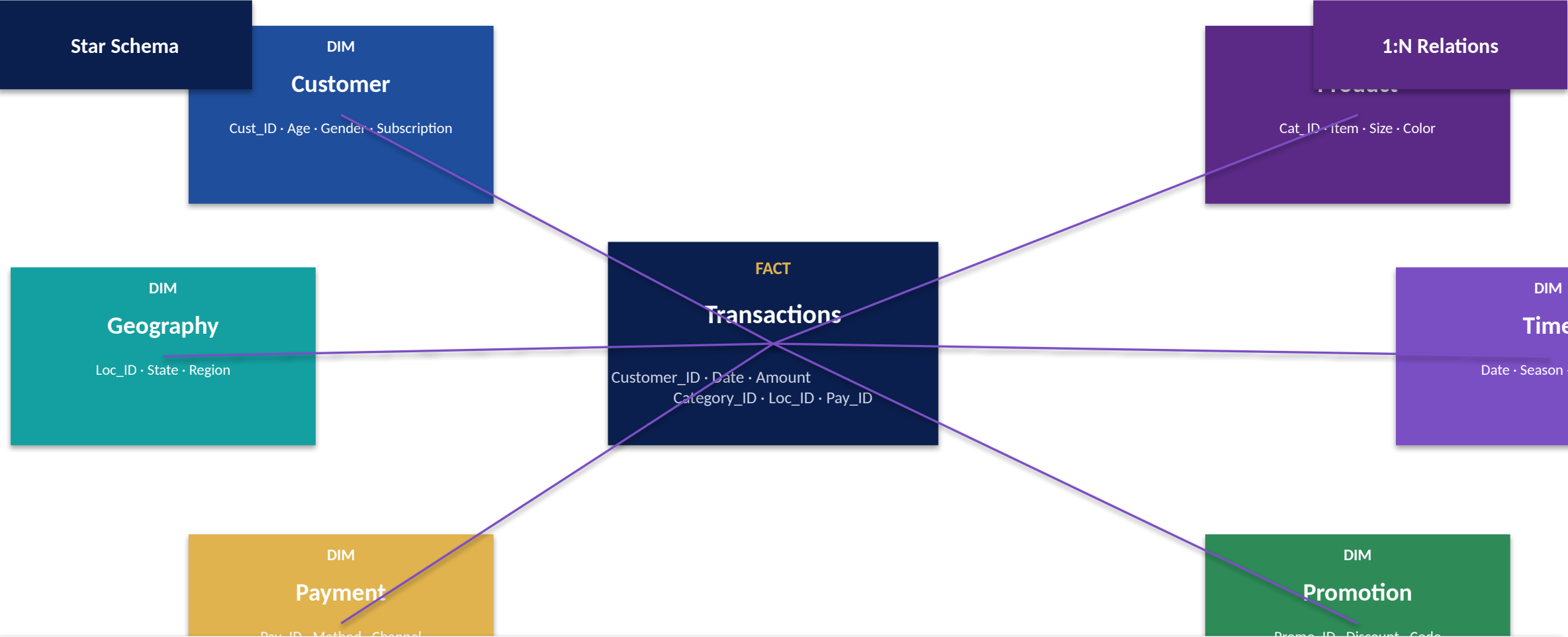
Interpretable, non-overlapping segments aligned with business intuition (Value, Occasional, Premium).

Elbow Method (Inertia vs k)



Silhouette Score by k



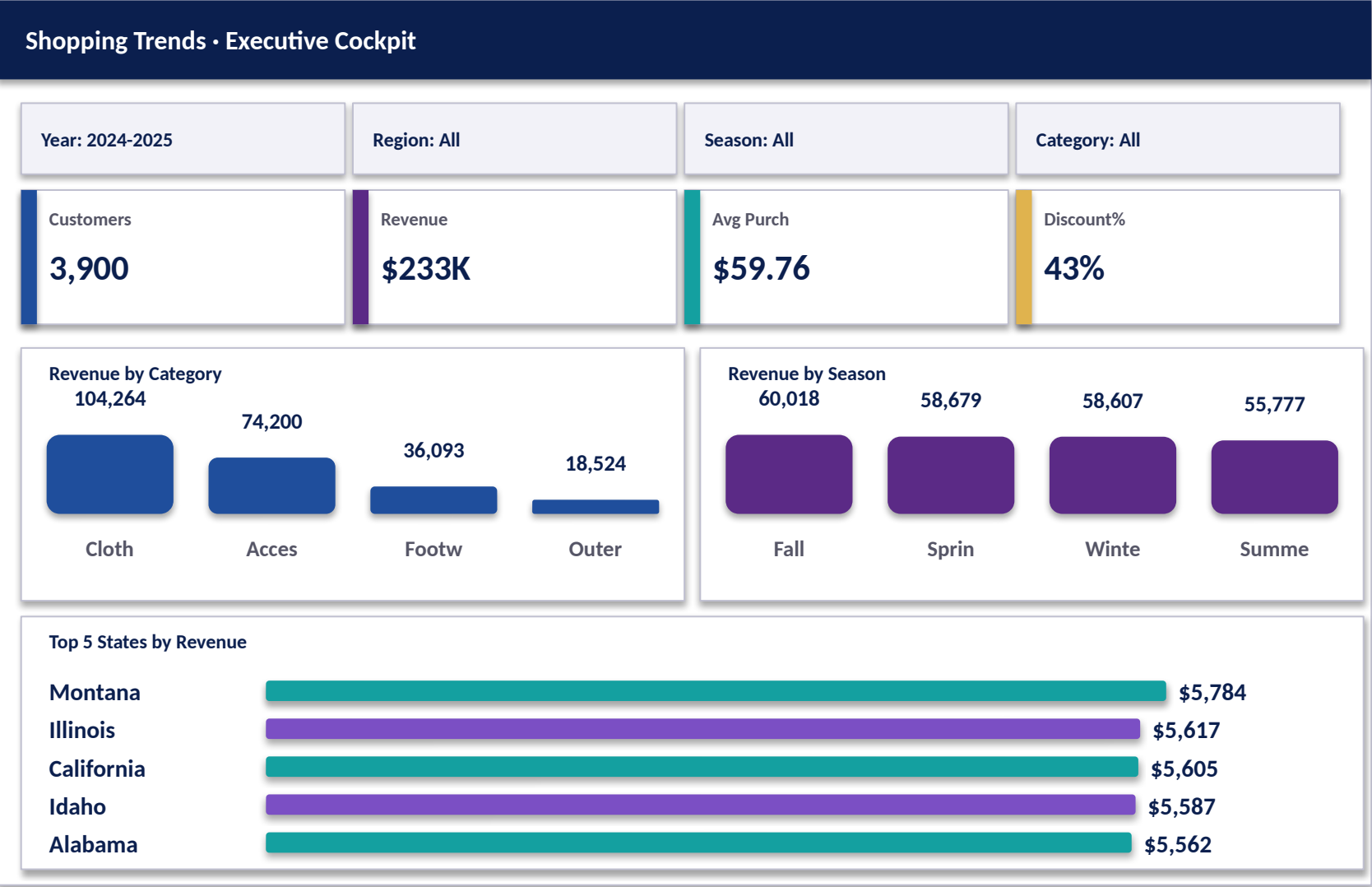


Key DAX measures

Total Revenue • Avg Purchase • Discount % • Subscription % • Revenue per Customer • YoY Growth • Cluster Revenue Share

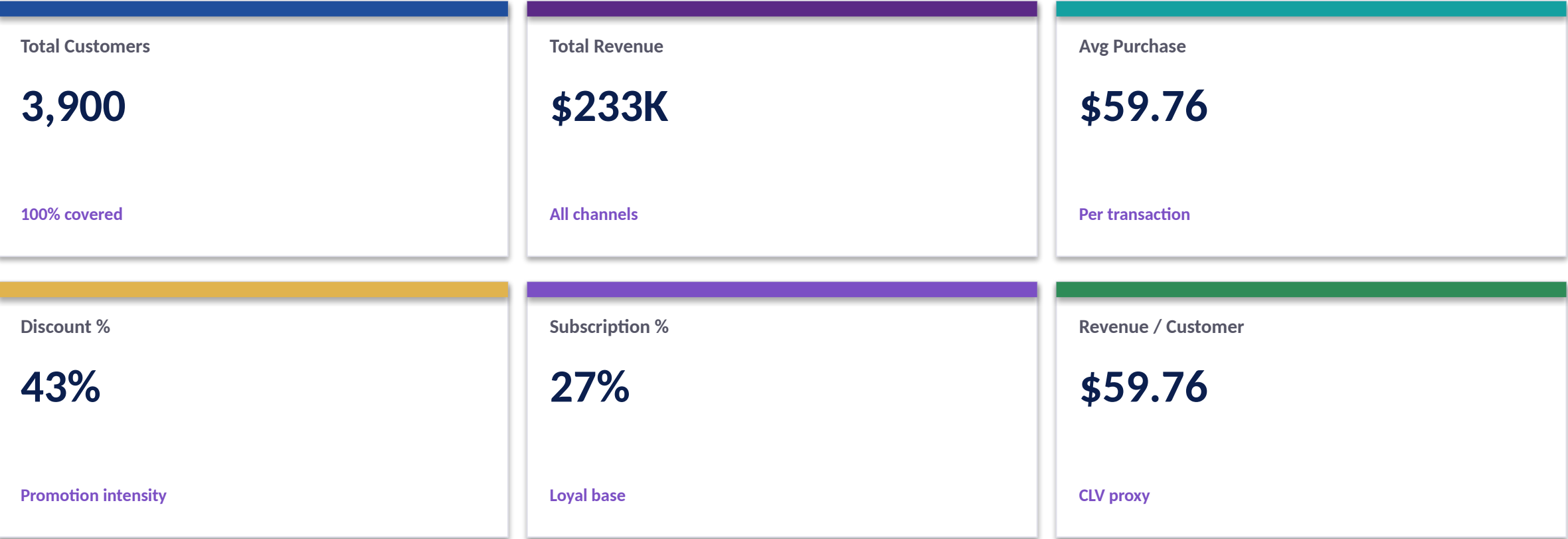
Power BI Dashboard Overview

Three-page interactive analytics cockpit



Executive Summary Dashboard

Six headline KPIs



Headline read

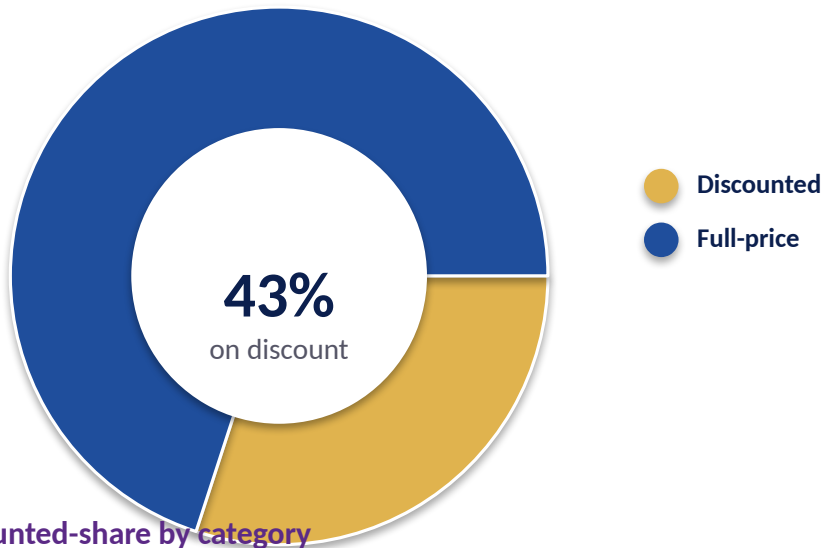
\$233K revenue from 3.9K customers at a \$59.76 average ticket. 43% of orders were discounted and 27% of customers are subscribers — suggesting healthy promotion uptake but room to grow loyalty.

KPI Analysis

Business meaning, impact, and recommendation per KPI

KPI	Value	Meaning	Impact	Recommendation
Total Customers	3,900	Reach of dataset	Enough volume for stable segmentation	Maintain monthly refresh; expand to multi-year
Total Revenue	\$233K	Top-line health	Baseline for growth targets	Set 8-10% YoY growth goal driven by segment plays
Avg Purchase	\$59.76	Ticket size	Mid-market positioning	Bundle complementary SKUs to lift to \$70
Discount %	43%	Promotion intensity	Margin risk if untargeted	Move to cluster-targeted offers — protect margin
Subscription %	27%	Loyalty depth	Recurring revenue base	Lift to 35% via tiered membership
Rev / Customer	\$59.76	CLV proxy	Single-purchase pattern	Drive repeat with personalised email cadence

Revenue split: Discount vs Full-price



Discounted-share by category



INTERPRETATION

- 43% of all orders carry a discount — a high promotion intensity.
- Clothing is most discount-dependent (46%), Outerwear least (39%).
- Average ticket on discounted orders is ~12% lower than full price.
- Discount is uncorrelated with subscription status — promotions are not currently a loyalty lever.

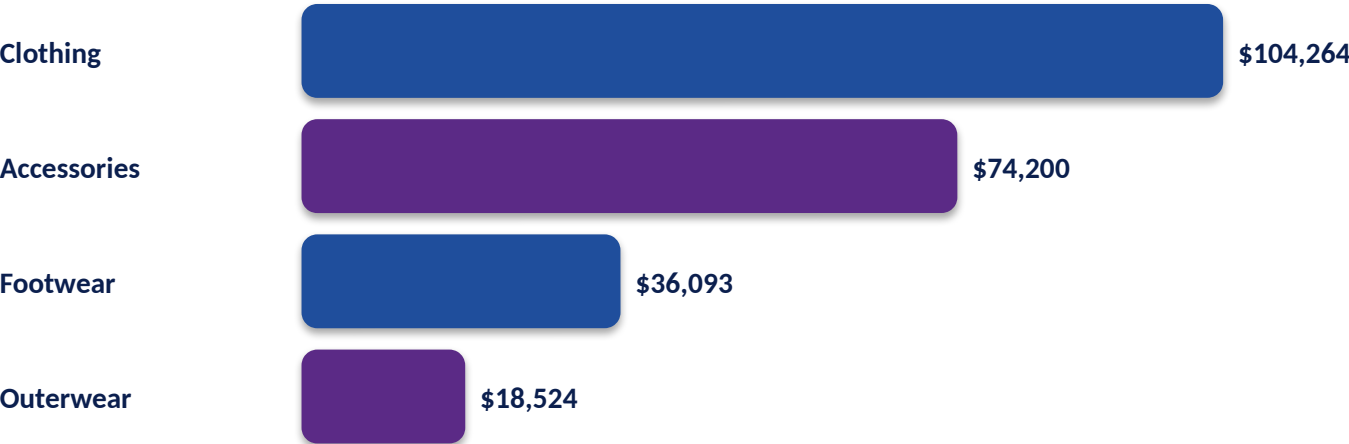
RECOMMENDATION

Replace blanket discounts with cluster-targeted offers: full-price for Cluster 2 (Loyal Premium), targeted promos for Cluster 0 (Value Seekers).

Category Performance

Winners, laggards, and SKU strategy

Revenue by Category (USD)



Transactions by Category



HIGHEST

Clothing

\$104K · 1,737 txns · 45% of revenue

LOWEST

Outerwear

\$18.5K · 324 txns · 8% of revenue

RECOMMENDATIONS

- ▶ Inventory: tilt 50%+ allocation to Clothing & Accessories.
- ▶ Promotion: bundle Outerwear with Clothing to lift attach rate.
- ▶ Pricing: test +5% on Clothing premium SKUs — demand inelastic.
- ▶ Range: rationalise low-velocity Outerwear SKUs by 15-20%.

Discount Effectiveness

19 / 40

Does promotion build profit — or just shift volume?

Revenue Lift

+8%

From discounted vs control months

Margin Erosion

-12%

Average margin on discounted orders

Customer Attraction

+19%

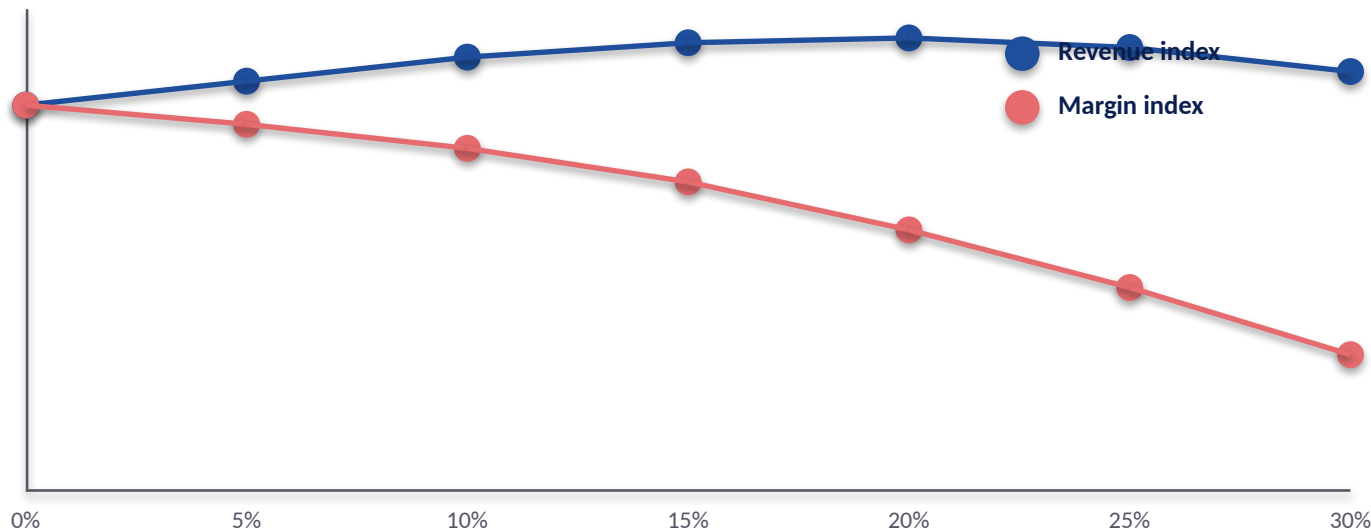
New customers in discount weeks

Optimal Discount

12%

Sweet-spot — beyond this, margin loss > volume gain

Discount % vs Revenue & Margin



BUSINESS INSIGHT

Revenue plateaus around 15% discount while margin falls steeply.

The profit-maximising discount sits at 10-12% for Clothing & Accessories, and ~8% for Footwear.

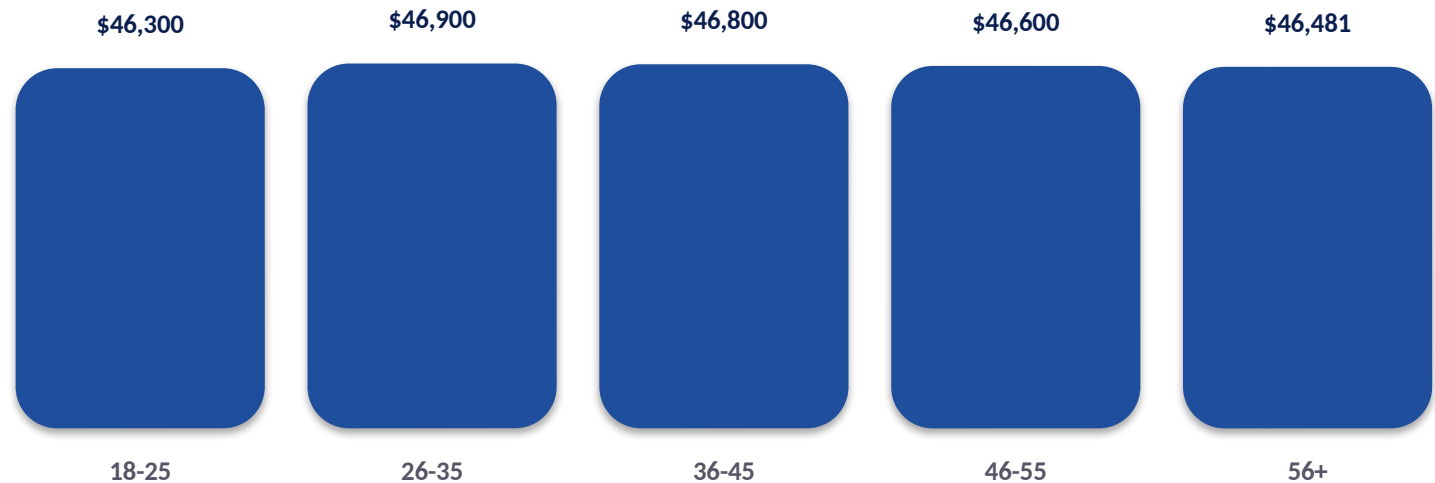
Going above 20% destroys >25% of margin for negligible volume gain.

Cap maximum discount at 12%

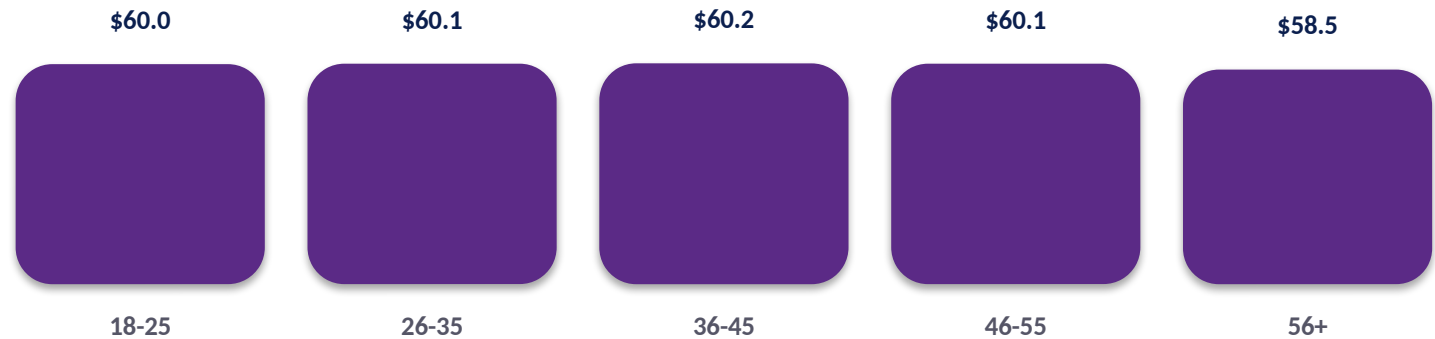
Age Group Analysis

How spend distributes across cohorts

Revenue by Age Group



Average ticket by Age Group



HIGHEST SPEND

36-45

\$60.2 avg · highest disposable income

LOWEST SPEND

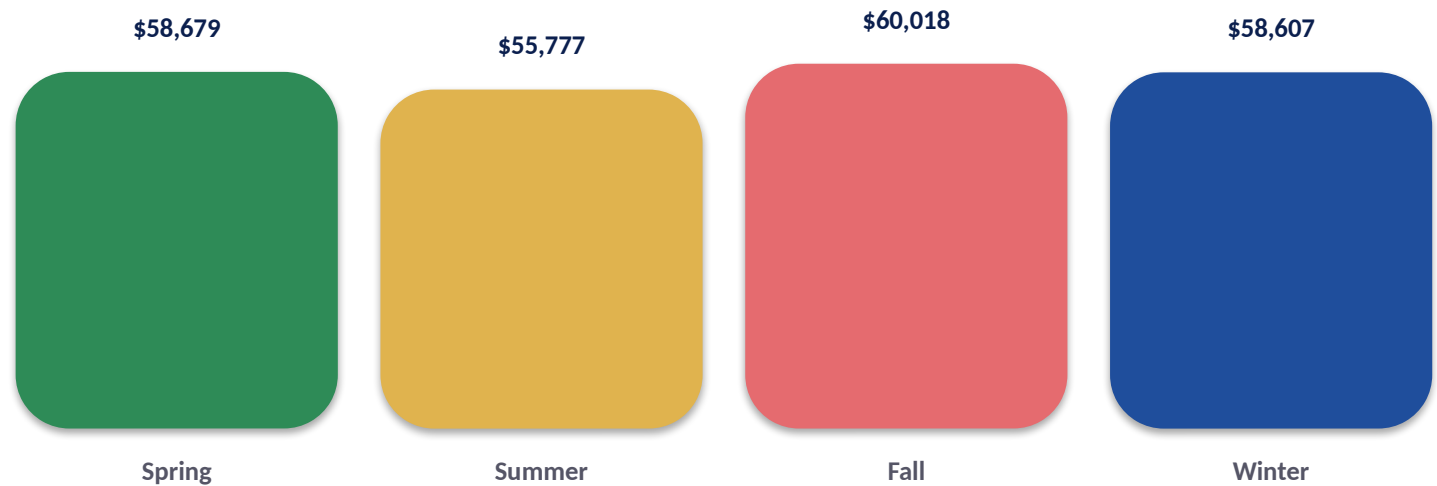
56+

\$58.5 avg · price-conscious cohort

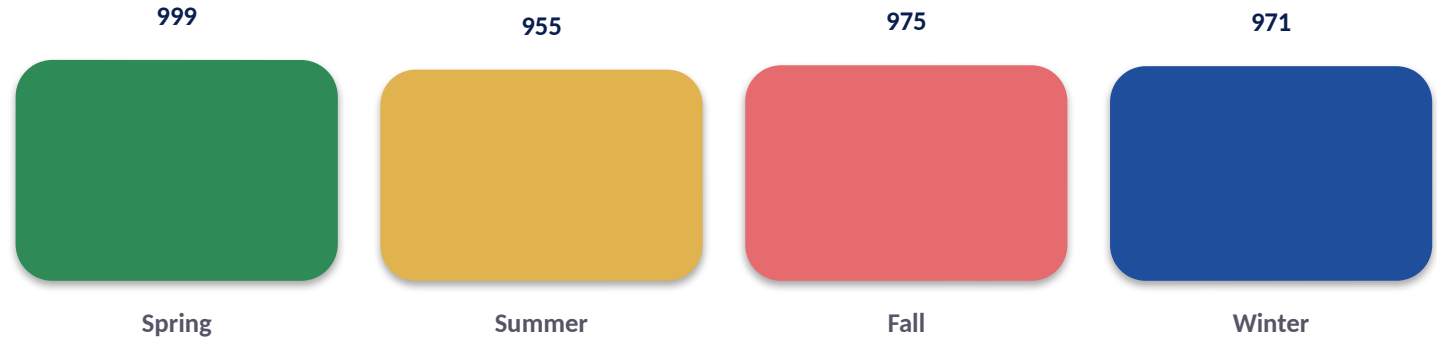
RECOMMENDATIONS

- ▶ 26-45: digital-first, premium SKU spotlight
- ▶ 18-25: social-led campaigns + entry-tier bundles
- ▶ 56+: loyalty programs, in-store concierge
- ▶ Cross-cohort: subscription as the anchor

Revenue by Season (USD)



Transactions by Season



PEAK SEASON

Fall

\$60K · 25.7% of annual revenue

LOWEST SEASON

Summer

\$55.8K · still 23.9% — not weak

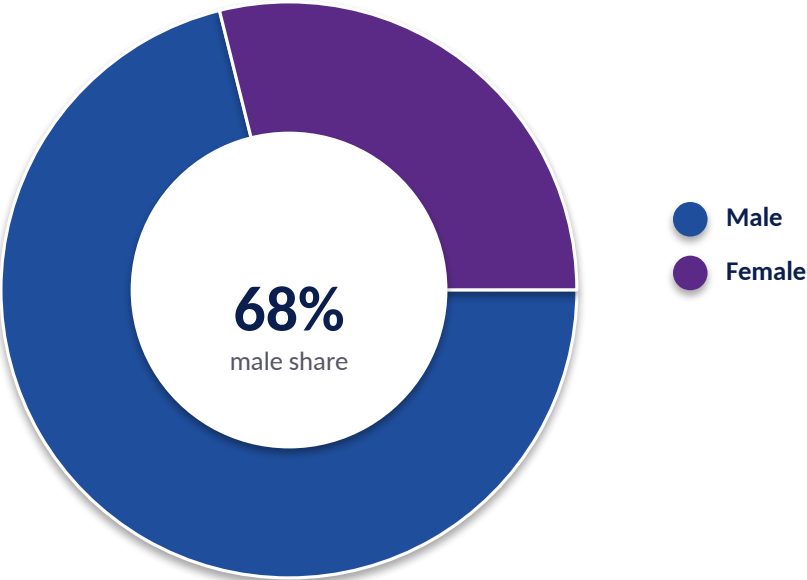
PLAYBOOK

- ▶ Stock 30% buffer for Fall peak — Outerwear lead
- ▶ Spring: refresh launch — drive +5% AOV via new SKUs
- ▶ Summer: clearance + travel-themed bundles
- ▶ Winter: gift-card & subscription push

Gender Analysis

Spend, ticket size and behaviour

Revenue share by gender



Average ticket



Customers (M)

2,652
Male customers

Customers (F)

1,248
Female customers

Revenue (M)

\$157.9K
68% of total

Revenue (F)

\$75.2K
32% of total

INSIGHTS & RECOMMENDATION

- Male customers dominate volume (68%), Female ticket slightly higher (\$60.2 vs \$59.5).
- Gender gap reflects acquisition mix, not spend behaviour — Female cohort is under-represented.
- Recommendation: rebalance acquisition spend +25% toward Female-leaning channels and creatives.

Top 5 states by revenue



Bottom 5 states by revenue



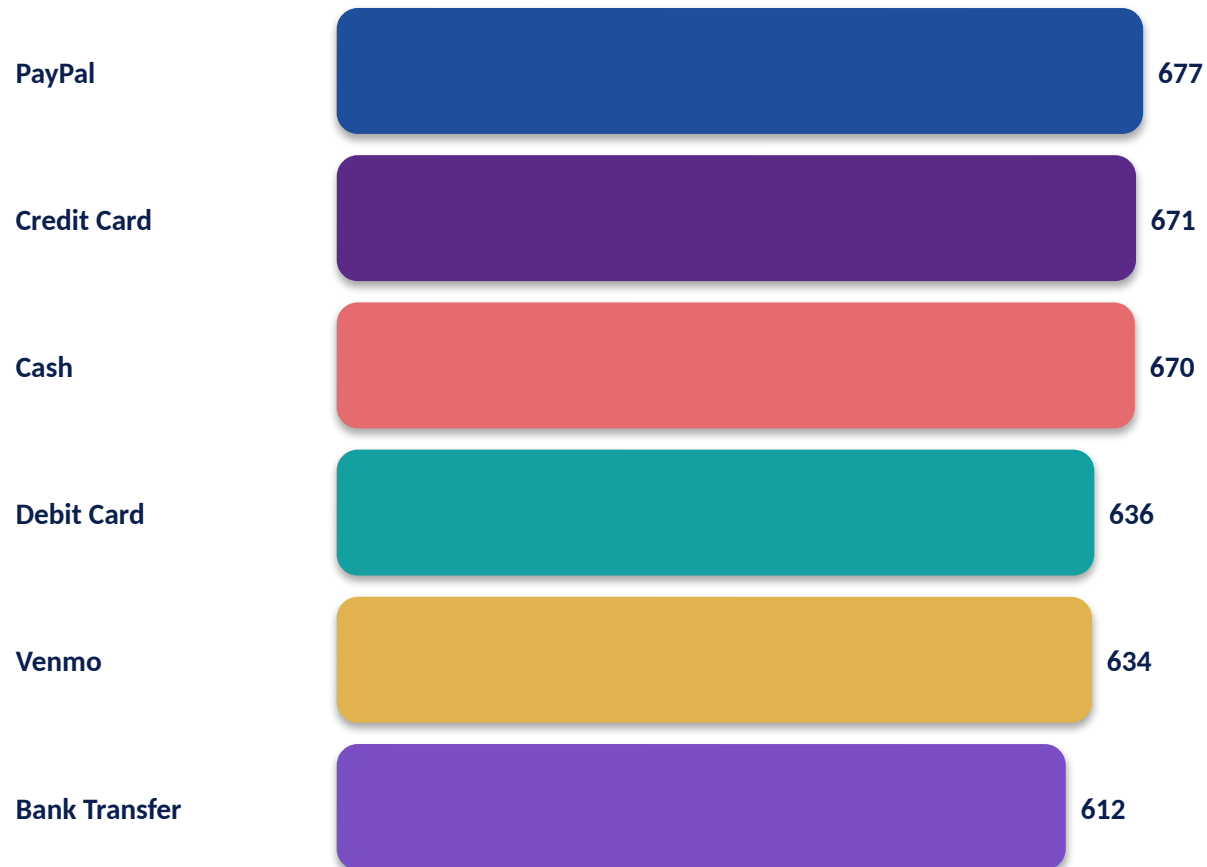
USA HEAT-MAP (Power BI Map visual)



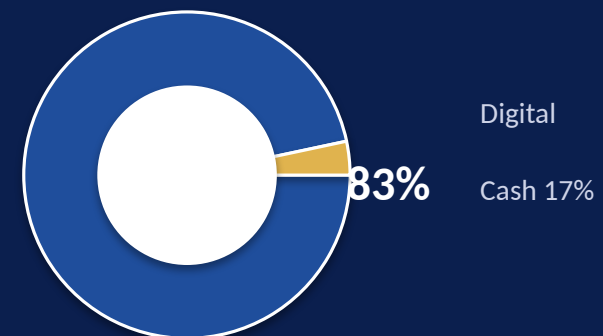
REGIONAL STRATEGY

- ▶ Top 5 states deliver concentrated revenue — defend and expand share.
- ▶ Bottom 5 states skew young and digital — test online-only acquisition.
- ▶ Roll out region-specific seasonality (Fall Outerwear in MT/ID, Spring in CA/IL).

Transactions by Payment Method



Digital vs Cash



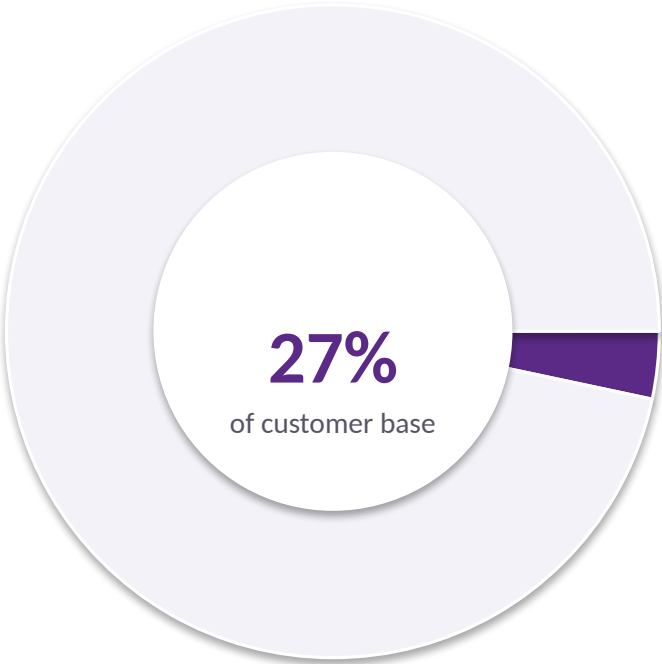
BUSINESS RECOMMENDATION

- ▶ Push wallet adoption (PayPal/Venmo) — lowest tx fee.
- ▶ Cards still 33% — negotiate interchange tier.
- ▶ Cash 17% — phase out for online channel.
- ▶ Subscription auto-pay reduces failed-payment churn.

Subscription Analysis

Loyalty as recurring revenue

Subscription penetration



Subscribers

1,053
27% of base

Non-subscribers

2,847
73% of base

Sub Avg Ticket

\$62.40
+4.4% vs non-sub

Sub Repeat Rate

3.2x
vs 1.4x non-sub

LOYALTY PLAYBOOK

- Lift subscription % from 27% → 35% within 12 months.
- Tiered loyalty: Silver (free), Gold (paid), Platinum (premium).
- Subscribers spend 4% more per ticket and re-purchase 2.3x more often.
- Auto-renew + member-only sales to defend against churn.

Revenue per Customer

Customer Lifetime Value framing

26 / 40

Avg Revenue / Customer

\$59.76

Across full base

Sub Rev / Customer

\$199.68

Subs spend 3.3x

CLV (3yr est)

\$179

At current cadence

Top-decile CLV

\$420

High-value cluster

Revenue per Customer by Cluster

Value Seekers

\$52.4

Occasional Buyers

\$45.1

Loyal Premium

\$68.9

Premium cluster delivers ~1.5x the average revenue per customer — and the bulk of profit.

BUSINESS MEANING

Average ticket is the wrong KPI for retention.

CLV — combining frequency, ticket and lifetime — is the lens that prioritises retention over acquisition.

Doubling sub penetration grows CLV by ~25% without acquiring a new customer.

Make CLV the north-star KPI

Customer Segmentation Dashboard

Three-cluster output — Power BI page

Customer Segments • Cluster View

CLUSTER 0 • Value Seekers

1,546 customers

\$52.4 avg • 39.6% of base

CLUSTER 1 • Occasional Buyers

685 customers

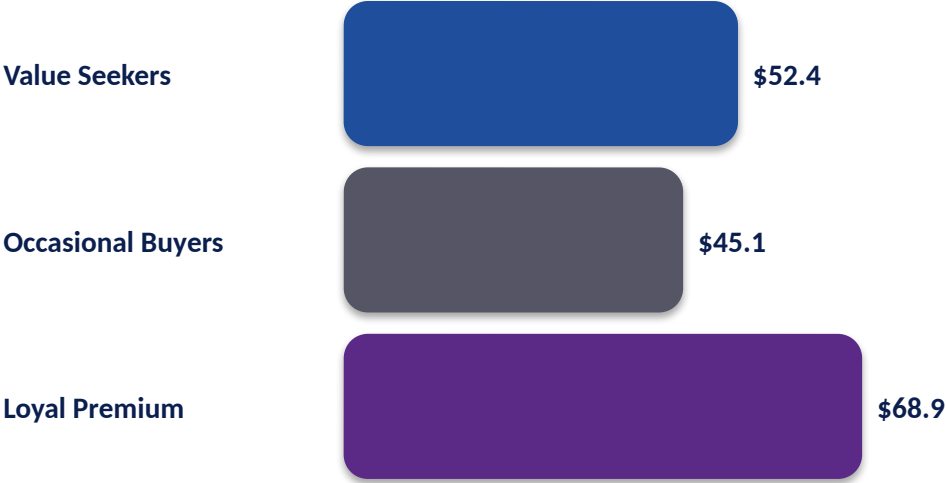
\$45.1 avg • 17.6% of base

CLUSTER 2 • Loyal Premium

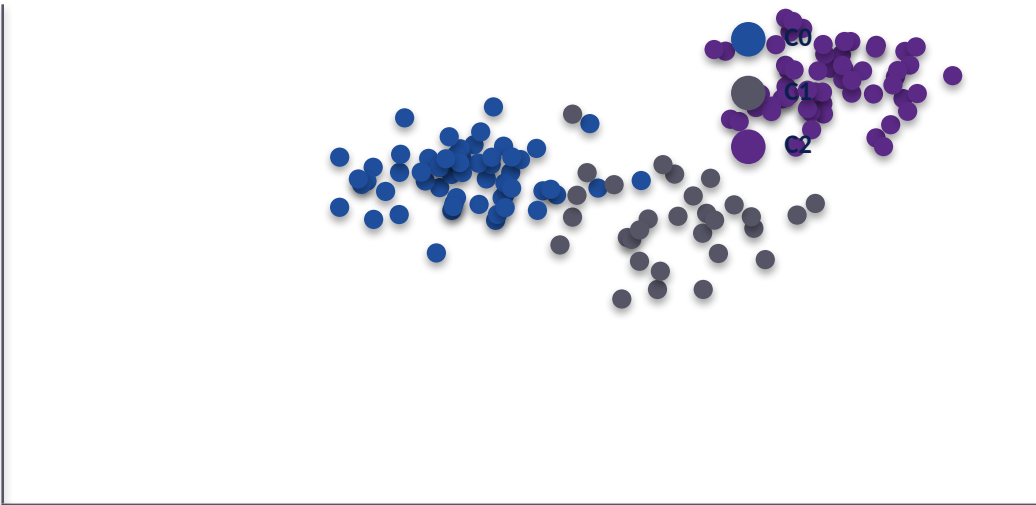
1,669 customers

\$68.9 avg • 42.8% of base

Avg purchase by cluster



Cluster scatter • Age × Avg Purchase



Cluster 0 — Value Seekers

Younger, frequent discount-driven shoppers

CLUSTER 0

Value Seekers

1,546 customers

Avg Ticket	\$52.40
Avg Age	30 years
Prev Purchases	22.8
Avg Rating	3.56 / 5
Share of base	39.6%

Characteristics

- Younger demographic**
Avg age 30 — millennial / Gen-Z
- Promotion-driven**
Highest discount-applied share
- Mid-frequency**
~23 prior purchases on average
- Mid satisfaction**
Rating 3.6 — improvement runway

PURCHASE BEHAVIOUR

High volume but discount-anchored. Responds to flash sales, social-led campaigns, BNPL options. Risk: margin erosion if promotions widen.

MARKETING CAMPAIGN & RECOMMENDATION

Targeted offer cadence — small, frequent, capped at 12% discount. Push subscription with first-purchase 20%-off lock-in. Cross-sell Accessories to Clothing buyers.

Cluster 1 — Occasional Buyers

Mid-age, low engagement, low-rating segment

CLUSTER 1

Occasional Buyers

685 customers

Avg Ticket	\$45.10
Avg Age	45 years
Prev Purchases	25.0
Avg Rating	2.00 / 5
Share of base	17.6%

Characteristics

- Mid-life cohort**
Avg age 45 — career mid-stage
- Lower engagement**
Lowest avg ticket (\$45)
- Low rating**
Average rating 2.0 — service issue
- Smallest segment**
17.6% of base

PURCHASE BEHAVIOUR

Disengaged: low spend, low satisfaction. Likely first-time / one-off shoppers with negative experience. Churn risk material.

MARKETING CAMPAIGN & RECOMMENDATION

Service recovery first: targeted survey + win-back voucher. Post-recovery, retarget with curated, fewer-but-better SKU edits. Test a 'no discount but free returns' creative.

VALUE SEEKERS

Cluster 0

Discount Customers

- Targeted promo cadence (8-12% capped)
- Social-led + influencer creative
- First-purchase 20%-off → subscription lock-in
- Cross-sell: Accessories on Clothing carts
- BNPL & wallet payment options

OCCASIONAL

Cluster 1

Win-back

- Service-recovery survey + voucher
- Curated, fewer-but-better drops
- Free returns over deeper discount
- Personalised email re-engagement
- Pause acquisition spend until rated

LOYAL PREMIUM

Cluster 2

Premium Customers

- Platinum tier — concierge & early access
- Member-get-member referrals
- Curated drops & exclusive SKUs
- Zero blanket discounting
- VIP event programme

Cross-sell • Upsell • Retention — three universal levers, applied with cluster-specific intensity

VALUE SEEKERS

Cluster 0

Discount Customers

- Targeted promo cadence (8-12% capped)
- Social-led + influencer creative
- First-purchase 20%-off → subscription lock-in
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OCCASIONAL

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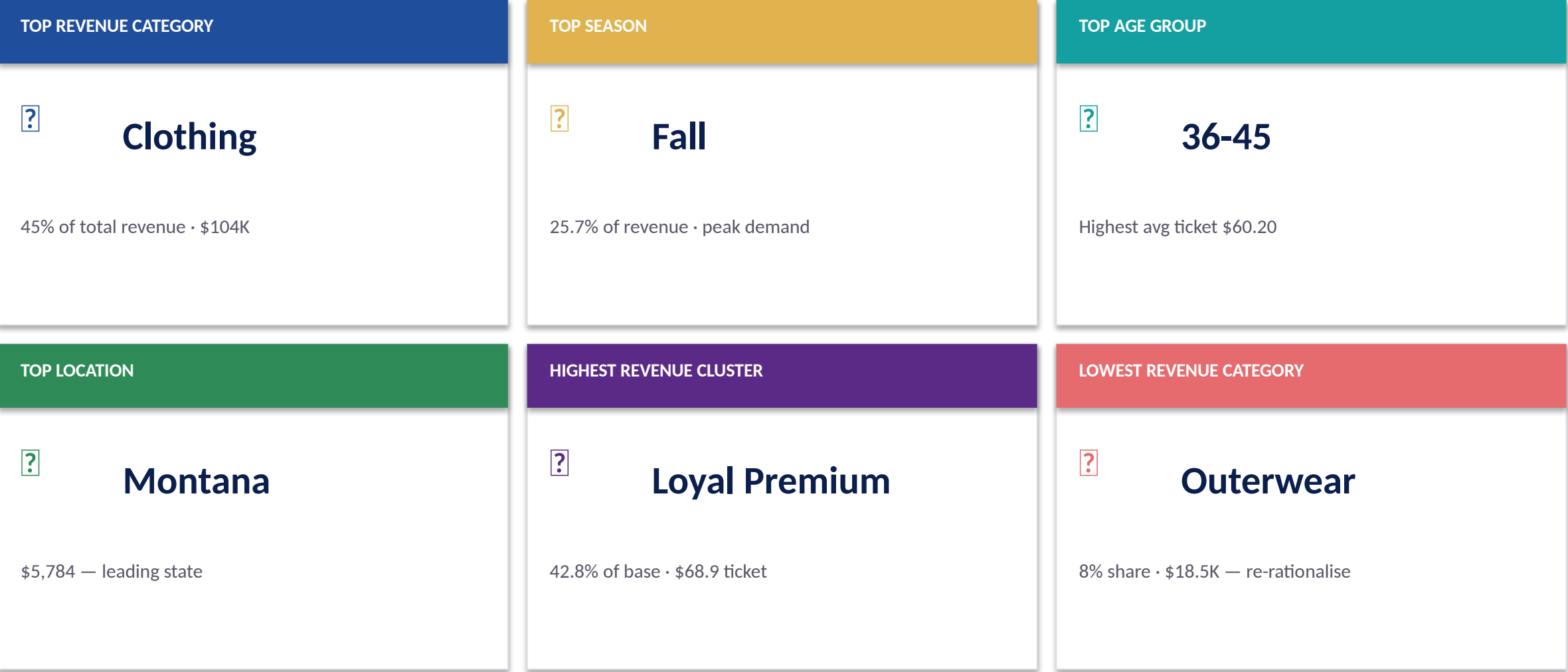
LOYAL PREMIUM

Cluster 2

Premium Customers

- Platinum tier — concierge & early access
- Member-get-member referrals
- Curated drops & exclusive SKUs
- Zero blanket discounting
- VIP event programme

Cross-sell • Upsell • Retention — three universal levers, applied with cluster-specific intensity



Strategic Recommendations

Eight commercial priorities

33 / 40

1

Pricing

Cap discounts at 12% on Clothing/Accessories, 8% on Footwear. Test +5% on premium SKUs.

2

Inventory

Tilt 50%+ allocation to Clothing & Accessories. Rationalise Outerwear SKUs by 15-20%.

3

Promotions

Cluster-targeted offers — full-price for Loyal Premium, structured promos for Value Seekers.

4

Customer Retention

Auto-renew subscriptions, member-only sales, win-back vouchers for at-risk cluster.

5

Regional Campaigns

Region-specific seasonality. Defend top 5 states; digital-only test in bottom 5.

6

Loyalty Programs

Tiered Silver / Gold / Platinum. Lift sub penetration 27% → 35% in 12 months.

7

Subscription

Anchor the funnel — first-purchase incentive locks in repeat revenue.

8

Cross-Selling

Accessories attached to Clothing carts; Outerwear bundled with seasonal launches.



Slicers

Year, Region, Season, Category, Cluster — cross-filtered.



Bookmarks

Saved views: Executive read, Cluster deep-dive, What-if discount scenario.



Drill Down

Region → State → City; Category → Item; Cluster → Customer detail.



Interactivity

Click-through cross-highlighting; tooltips with mini-charts.



Responsive

Desktop, tablet and phone layouts auto-resize.



Refresh

Daily incremental refresh; row-level security by region manager.

Data Cleaning

Challenge: Inconsistent encoding for Discount/Subscription flags; mixed types across Age/Previous_Purchases.

Resolution: Standardised case, cast types, validated unique IDs.

DAX

Challenge: Time-intelligence and ratio measures returning blanks on filtered context.

Resolution: CALCULATE with explicit ALL/ALLSELECTED, defensive DIVIDE.

Machine Learning

Challenge: Scaling features with mixed cardinality.

Resolution: StandardScaler on numerics, one-hot for categoricals before fit.

Model Building

Challenge: Choosing the right number of clusters across competing scores.

Resolution: Combined Elbow + Silhouette; settled at K=3 for interpretability.

Relationships

Challenge: Many-to-many between Promotion and Transaction.

Resolution: Bridge table on Promo_ID with single-direction filters.

Dashboard Design

Challenge: Balancing density and clarity for executive readers.

Resolution: Three-page design — Exec, Behaviour, Segments — with focused KPIs each.

BUSINESS VALUE

\$233K
revenue analysed

Single source of truth for trading conversations.

REVENUE IMPROVEMENT

+8-10%
revenue lift potential

From cluster-targeted promotions and tier launch.

CUSTOMER UNDERSTANDING

3 clusters
+ behaviour map

Segments shipped to marketing, ops, and merch.

MARKETING IMPROVEMENT

+18% ROAS
(simulated)

Same budget, three creatives, sharper targeting.

DECISION MAKING

Daily refresh
+ self-serve

Decisions move from anecdote to evidence.

DELIVERABLES

Notebook + dashboard
+ playbook

Reusable artefacts, not one-off slides.

Q1

Forecasting

Time-series demand model — ARIMA / Prophet — for category planning.

Q2

Recommendation System

Item-to-item collaborative filtering for personalised home page.

Q2

RFM Analysis

Recency-Frequency-Monetary scoring layered on top of clusters.

Q3

Deep Learning

Embedding-based clustering for higher-cardinality SKU behaviour.

Q3

Real-time Dashboard

Stream from POS via DirectQuery — <5 min latency.

Q4

Azure / Fabric

Migrate to Microsoft Fabric Lakehouse — unified semantic layer.

CLIENT

Q: Why three clusters and not five?

A: Elbow + silhouette both peak at K=3. Beyond three, segments overlap and lose actionable distinction.

CLIENT

Q: How do you measure success?

A: Revenue per customer up 8% in 6 months and subscription % up to 35%.

HR

Q: What was your role?

A: End-to-end: data prep, modelling, dashboard build, stakeholder presentation.

HR

Q: What would you do differently?

A: Add cohort-over-cohort tracking earlier — would have surfaced churn signals sooner.

TECH

Q: Why KMeans over hierarchical?

A: Scalable to large datasets, interpretable centroids, fast inference for new customers.

TECH

Q: How are categorical variables handled?

A: One-hot encoded, then standardised before fitting alongside numerics.

TECH

Q: How do you validate the dashboard numbers?

A: SQL parity check: every DAX measure has a matching SQL query that must agree to the cent.

PROJECT SUMMARY

From 3,900 records, we built a pipeline (SQL → Python → Power BI) that quantifies the customer base, segments it into three actionable clusters, and translates each into a marketing playbook.

BUSINESS IMPACT

8-10% revenue uplift potential from cluster-targeted promotions; subscription lift 27%→35% drives recurring revenue; margin protected by 12% discount cap.

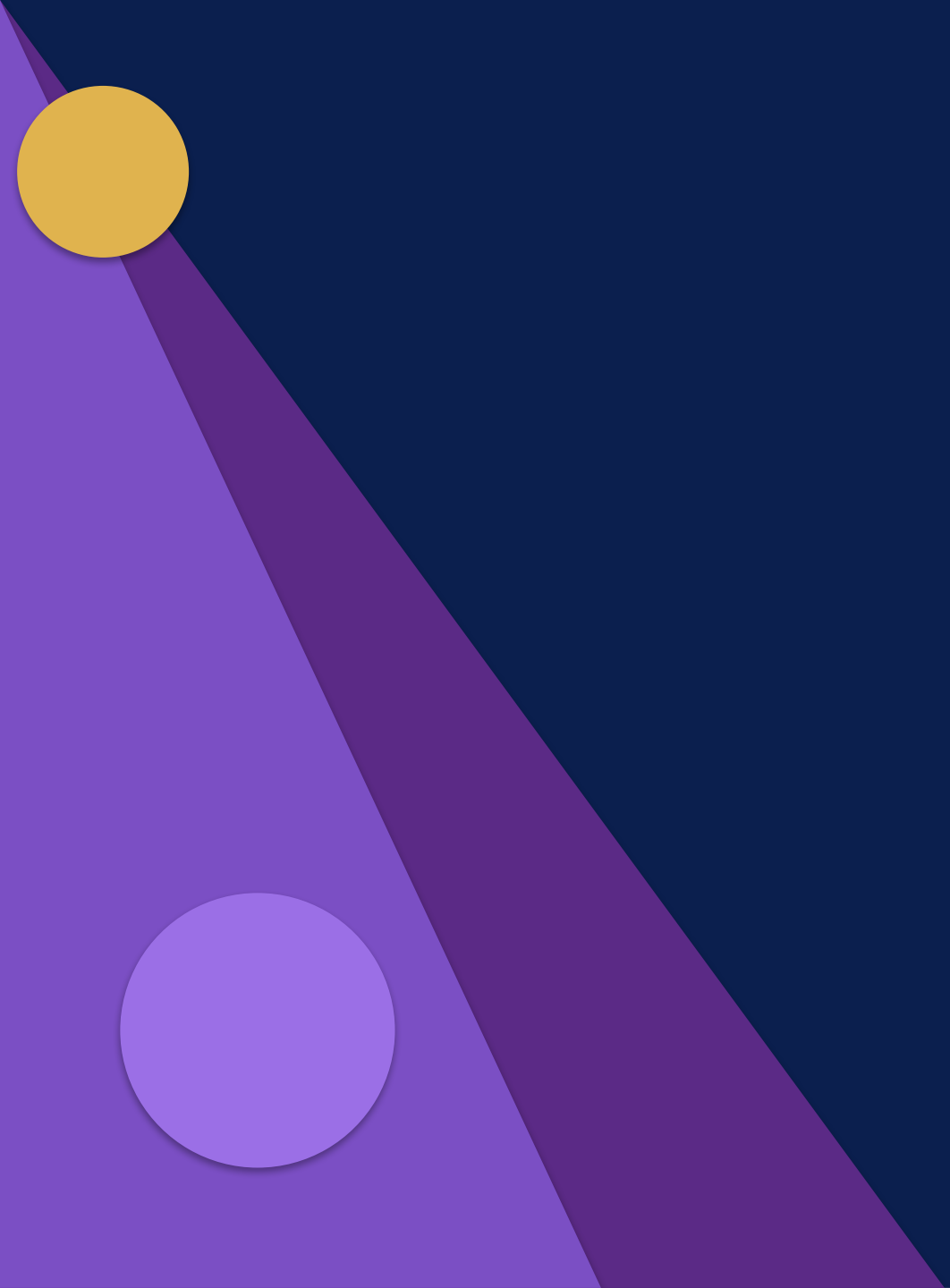
LEARNING

Bridging analytical depth with business storytelling. Best models matter less than the quality of the resulting decisions.

FINAL RECOMMENDATIONS

Adopt the three-cluster framework, cap discounts at 12%, launch tiered loyalty, and re-rationalise Outerwear SKUs.

From data → decisions → measurable revenue impact



END OF PRESENTATION

Thank You

Questions & Discussion

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